

SHOT3R: MULTI-SHOT STATE TRANSITIONS FOR STREAMING 4D HUMAN-SCENE RECONSTRUCTION

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Figure 1: Shot3R streams successive monocular shots into a common world frame while preserving camera, motion, and identity continuity. Angles denote source-camera rotations from the first shot.

ABSTRACT

Streaming multi-person 4D reconstruction from monocular videos with shot transitions requires camera trajectories, humans, and scene geometry from successive shots to share a common world frame. Existing online recurrent methods update this reconstruction incrementally by carrying a recurrent state forward, an operation that generally assumes visual and geometric continuity between adjacent observations. At a shot boundary, however, the preceding state has conflicting roles: it provides a spatial reference for relating the new shot to the accumulated reconstruction, but becomes unreliable as a recurrent condition within the new shot. Carrying this state forward can corrupt new-shot estimates, whereas resetting it discards the cross-shot spatial relation. **Shot3R** resolves this conflict by formulating each shot transition as a history-conditioned online state transition. At each boundary, Shot3R jointly models the historical representation and the current observation in latent space to infer the spatial relation between the new shot and the existing reconstruction, while initializing an independent recurrent state for the new shot. This boundary-time inference establishes a consistent world-frame representation of camera trajectories, human motion, and scene geometry; geometry-based person association preserves identities across shots, and only the new-shot state is propagated thereafter. In controlled comparisons across three multi-person benchmarks, Shot3R reduces the dataset-balanced mean W-MPJPE by 9.5% and improves mean IDF1 by 0.099 in absolute terms relative to continuous state propagation. On EgoBody’s predefined extreme-viewpoint subset, it reduces W-MPJPE by 29.9%. Comparisons with publicly available online and offline reconstruction systems under the same multi-shot inputs further show competitive reconstruction accuracy.

1 INTRODUCTION

Recent advances in streaming 4D reconstruction have made it possible to jointly recover human motion, camera trajectories, and scene geometry from a monocular video as it is processed sequentially (Wang et al., 2025b; Chen et al., 2026b; Sur et al., 2026). These methods achieve efficient online inference by compressing past observations into a recurrent state, but their formulation largely assumes that adjacent frames belong to a continuous capture. Edited videos routinely violate this assumption: they comprise temporally successive shots separated by transitions that may abruptly change viewpoint, remove visual overlap, and alter the visible scene and human appearance (Pavlakos et al., 2022b; Zhang et al., 2025b; On et al., 2026). The history accumulated before such a boundary may therefore no longer provide valid recurrent context for the new shot. We study streaming reconstruction under these discontinuities. Here, *multi-shot* refers to successive segments of a monocular video, rather than synchronized multi-view observations. The goal is to express the camera trajectories, reconstructed humans, and scene geometry of every shot in a *common world frame*—the reconstruction coordinate system initialized by the first shot and shared across the sequence.

Existing methods address only one side of this setting. Multi-shot human reconstruction approaches relate people across transitions by reasoning over multiple available shots, often through joint processing or sequence-level optimization (Pavlakos et al., 2022b; Zhang et al., 2025b; On et al., 2026). Although effective for recovering inter-shot structure, they are generally not designed to produce each estimate from a bounded recurrent state without future-frame access. Stateful online reconstruction methods provide the complementary capability: they process frames in temporal order and avoid per-video global optimization (Wang et al., 2025b; Chen et al., 2026b; 2025). Their recurrent state, however, is designed to evolve through continuous observations and provides no explicit mechanism for a sudden change in history validity at a shot boundary. Neither line of work therefore addresses the central requirement here: preserving the spatial reference established by earlier shots while preventing incompatible history from being propagated through reconstruction of the new shot.

This gap arises from a conflict in how recurrent history is used at a shot boundary. Carrying the state across the boundary preserves the coordinate reference established by previous observations, but forces the new shot to interact with history accumulated under a discontinuous viewpoint; errors introduced at the transition can then persist through subsequent recurrent updates. Reinitializing the state removes this interference and restores coherent recurrence within the new shot, but also erases the reference needed to locate that shot in the common world frame. The key question is therefore not whether history should be retained or discarded, but which of its roles should survive the boundary. Our central insight is that history should survive only as a temporary alignment reference: it may be read to establish the inter-shot relation, but should not be propagated as the recurrent state of the new shot. A controlled comparison using the same reconstruction model supports this distinction: state carry-over produces time-varying camera drift rather than a single fixed offset, whereas independent reconstruction remains coherent within the new shot but loses its relation to the preceding world frame (Section 4.3).

To resolve this conflict, we introduce Shot3R, a streaming reconstruction framework that assigns different lifetimes to the two roles of recurrent history. At each shot boundary, Shot3R temporarily consults the preceding state to infer the inter-shot relation, without retaining the resulting state as memory for subsequent reconstruction. Recurrence within the new shot instead starts from an independently initialized state, which becomes the only state propagated forward. Once the spatial relation is established, geometry-based person association maintains identities across the boundary, and a shared camera-human transform expresses the new-shot reconstruction in the common world frame. Shot3R operates without future-frame access, revision of previously emitted estimates, or per-video optimization. Across three multi-person benchmarks, it reduces the dataset-balanced mean W-MPJPE by 9.5% over continuous state propagation and consistently improves camera accuracy and identity continuity; on EgoBody’s predefined extreme-viewpoint subset, the W-MPJPE reduction reaches 29.9%.

Our contributions are:

- We identify and characterize a conflict in recurrent history at shot boundaries: history is necessary for relating successive shots, yet propagating it into the new shot can introduce

persistent reconstruction errors, while discarding it entirely removes the inter-shot spatial reference.

- We introduce Shot3R, a streaming multi-shot reconstruction framework that separates how history is read for alignment from how state is propagated for reconstruction. Shot3R retains pre-boundary history only as a temporary reference and advances the new shot using independently initialized recurrence, while maintaining spatial and identity consistency across the boundary.
- Through controlled same-model comparisons and evaluations on three multi-person benchmarks, we show that this separation improves world-frame human reconstruction, camera accuracy, and identity continuity, with its largest gains under substantial viewpoint changes and without future-frame access or per-video optimization.

2 RELATED WORK

Global human motion reconstruction. Human mesh recovery has evolved from fitting parametric body models to feed-forward regression from monocular images and videos (Kanazawa et al., 2018; Kocabas et al., 2020; Goel et al., 2023). Image-based methods primarily estimate human pose and shape in the camera frame, while temporal models and full-frame multi-person approaches improve motion consistency and reasoning in crowded scenes (Sun et al., 2021b; 2022; 2023; Baradel et al., 2024; Wang et al., 2025c; Sun et al., 2024). When the camera itself moves, however, camera-frame predictions do not directly describe global human motion in the surrounding 3D environment. Existing methods therefore recover human trajectories together with camera motion through motion priors and sequence optimization (Yuan et al., 2022; Ye et al., 2023; Kocabas et al., 2024; Li et al., 2024), joint human-camera modeling (Wang et al., 2024b; Shin et al., 2024; Shen et al., 2024; Yin et al., 2024), or perspective and scale calibration (Patel & Black, 2025; Yang et al., 2025). More recent systems further incorporate humans, cameras, and surrounding scene geometry into a unified reconstruction (Müller et al., 2025; Liu et al., 2026b; Zhang et al., 2025c; Li et al., 2026a; Rojas et al., 2025; Yang et al., 2026; Liu et al., 2026a). Offline methods can exploit global constraints from a complete sequence, but typically require iterative joint optimization of human motion, camera trajectories, and scene geometry over the entire video, incurring substantial latency and computation and limiting scalability to growing sequences. Unified feed-forward methods reduce the cost of per-sequence optimization; continuous video streams additionally require reconstruction to be updated online using only current and past observations, with a bounded history representation and stable per-step computation. The question thus shifts from recovering global relationships from a complete video to maintaining them through an evolving history representation during online inference.

Online reconstruction. To this end, online reconstruction processes video in temporal order and continually updates estimates of cameras, geometry, and people. Classical SLAM, learned dense mapping, and feed-forward point-map models provide the foundations for incremental camera localization, scene mapping, and direct 3D prediction, respectively (Teed & Deng, 2021; Sun et al., 2021a; Wang et al., 2024a; Leroy et al., 2024; Zhang et al., 2025a; Wang et al., 2025a). Recent streaming methods further preserve past observations through spatial memory, progressive registration, or a recurrent state (Wang & Agapito, 2025; Liu et al., 2025; Wang et al., 2025b). For human reconstruction, OnlineHMR studies online recovery in world coordinates (Zhao et al., 2026), while Human3R builds on CUT3R’s recurrent reconstruction to jointly recover cameras, scenes, and multiple people online (Chen et al., 2026b). Building on this line, UniCon3R and GUSH3R extend unified reconstruction through human-scene contact and renderable dynamic representations, respectively (Sur et al., 2026; Abe et al., 2026); other streaming 3D methods study long-term memory, dynamic geometry, and stable recurrent updates (Chen et al., 2025; Wu et al., 2025; Zhuo et al., 2025; Chen et al., 2026a; Zheng et al., 2026; Li et al., 2026b). Existing online methods, however, are primarily designed for continuously captured monocular video and typically rely on stable temporal relations and visual overlap between adjacent frames to carry the reconstruction forward. A shot transition abruptly changes the camera viewpoint and visible content, invalidating this continuity assumption even though reconstructions from different shots must still share a coherent spatial relation. Extending online reconstruction to multi-shot video therefore requires consistency to be maintained across shot transitions.

Multi-shot human reconstruction. Multi-shot human reconstruction aims to recover spatially consistent multi-person 3D motion from monocular videos with shot transitions while preserving identities across shots. Early work separately studied person tracking and re-identification in TV content, single-person recovery across shots, and joint actor–environment reconstruction (Pavlakos et al., 2022a;b;c). HumanMM aligns human orientation and global trajectories across shots through camera-motion estimation and sequence-level motion integration, but focuses on single-person videos (Zhang et al., 2025b). Most closely related to our setting, Multi-THuMBS uses a 3D scene prior to reconstruct observations on both sides of a shot boundary in a common space, associates people using appearance, pose, and geometric cues, and then optimizes the global result with trajectory smoothing and a cross-camera reprojection constraint (On et al., 2026). It extends multi-shot reconstruction to multiple people, but adopts multi-stage global optimization over an observed sequence. ShowMak3R also handles multi-person, multi-shot content, but optimizes per-video neural representations of actors and scenes, targeting renderable cinematic reconstruction rather than online human-motion recovery (Kim et al., 2025). Cross-shot identity persistence is further related to 3D-aware tracking and appearance-, pose-, or keypoint-based re-identification (Rajasegaran et al., 2022; He et al., 2021; Yuan et al., 2025; Somers et al., 2024). Together, these works advance multi-shot human reconstruction through global motion recovery, spatial alignment, and person association, but commonly jointly process observations across the shot boundary or optimize a complete sequence. They therefore leave open how a maintained history representation should cross a shot boundary during online recurrent reconstruction. Shot3R addresses this streaming setting by using history temporarily to establish the inter-shot spatial relation, without propagating it as the recurrent state of the new shot, thereby maintaining spatial and identity consistency under sequential multi-shot processing.

3 METHOD

Shot3R processes a multi-shot monocular video in temporal order and reconstructs its cameras, visible people, and scene geometry in a common world frame. Within each shot, a recurrent reconstructor continually updates a compact history representation. At a shot transition, this history remains a useful reference for establishing the inter-shot spatial relation, but is no longer a reliable recurrent condition for the new visual content. Shot3R therefore assigns different information lifetimes to these two roles of history.

As illustrated in Figure 2, a detected boundary triggers two complementary evaluations of the same first new-shot frame. A history-conditioned alignment path temporarily reads the preceding state to infer the inter-shot relation and is then discarded. A reinitialized reconstruction path starts the new shot independently and owns the only state propagated forward. Typed alignment representations organize boundary evidence from the current observation and recurrent history. The resulting relation, geometry-based person association, and a shared transform then commit the reinitialized outputs to the existing world frame. This boundary operation uses only current and past observations, revises no previously emitted frame, and requires no per-video optimization.

3.1 RECURRENT RECONSTRUCTION AT SHOT BOUNDARIES

Given a monocular stream $\{I_t\}_{t=1}^T$ composed of consecutive video shots, Shot3R uses only observations up to time t to recover camera pose C_t , scene geometry P_t^w , and multiple visible people $\mathcal{H}_t$ online. All outputs are expressed in a *common world frame*: the reconstruction coordinate system initialized by the first shot and shared by the cameras, people, and scene geometry of every subsequent shot, rather than an externally specified absolute coordinate system. Encoding past observations in a recurrent state S_t , a general stateful reconstruction process is

$$(y_t, S_t) = f_\theta(I_t, S_{t-1}), \quad y_t = \left(C_t, P_t^w, \{J_t^i, V_t^i, q_t^i\}_{i=1}^{N_t} \right), \quad (1)$$

where C_t is a camera-to-world pose, and J_t^i , V_t^i , and q_t^i denote the SMPL-X (Pavlakos et al., 2019) joints, mesh vertices, and anonymous slot of person i , respectively. We instantiate f_θ with a pretrained Human3R recurrent reconstructor, which supplies within-shot camera, human, and scene predictions (Chen et al., 2026b). Under continuous observations, the spatial-reference and recurrent-conditioning roles of S_{t-1} are compatible. A shot transition separates them: the preceding state still

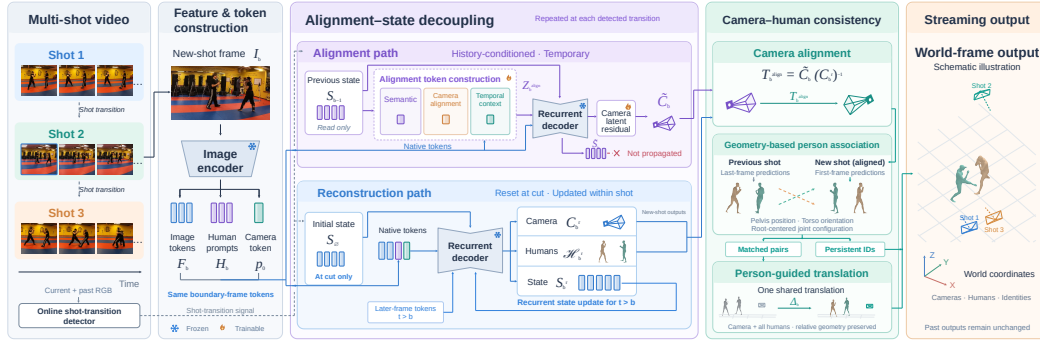


Figure 2: Overview of Shot3R. The history-conditioned path temporarily infers the inter-shot relation, while only the reinitialized path propagates recurrent state. Geometry-based association transfers identities, and a shared transform places new-shot outputs in the common world frame. The rightmost panel is schematic.

relates the new observation to the established reconstruction, but its viewpoint- and content-specific representation is no longer a valid condition for updating the new-shot state. Directly carrying it forward preserves the reference but can propagate this mismatch; reinitializing it isolates the mismatch but removes the inter-shot relation.

To resolve this conflict, Shot3R performs two complementary online evaluations at a detected boundary b :

$$\begin{aligned} (\tilde{y}_b, \tilde{S}_b) &= f_{\theta, \phi}(I_b, S_{b-1}), & (y_b^r, S_b^r) &= f_{\theta}(I_b, S_{\emptyset}), \\ (y_t^r, S_t^r) &= f_{\theta}(I_t, S_{t-1}^r), & t &> b. \end{aligned} \quad (2)$$

The history-conditioned alignment path reads S_{b-1} only to infer the relation between the new shot and the existing world frame; its predictions and $\tilde{S}_b$ are discarded after the boundary operation. The reinitialized reconstruction path starts from $S_{\emptyset}$, and S_b^r becomes the sole recurrent state for all subsequent frames in the shot. Thus, inter-shot relation estimation and new-shot state propagation have separate information paths.

3.2 HISTORY-CONDITIONED SHOT ALIGNMENT

The history-conditioned path must estimate an inter-shot relation from the interaction between the first new-shot frame I_b and the preceding state S_{b-1} . Recurrent features designed for continuous reconstruction mix image content, camera motion, human information, and historical context, whereas boundary reasoning benefits from keeping their different roles identifiable. Rather than prematurely collapsing them into a single correction vector, Shot3R explicitly constructs three typed alignment representations for semantic correspondence, camera-latent change, and accumulated temporal context.

Let c_b denote observation evidence pooled from the current image, camera, and human tokens, and let m_{b-1} denote historical evidence retrieved from the recurrent state and camera memory. The typed representations are

$$\begin{aligned} z_b^{\text{sem}} &= \alpha_b c_b + (1 - \alpha_b) m_{b-1}, & z_b^{\text{cam}} &= \text{MLP}_{\text{cam}}[p_b, p_{b-1}, \Delta p_b, m_{b-1}], \\ z_b^{\text{temp}} &= \text{MLP}_{\text{temp}}[\bar{z}_{b-1}, \delta_{b-1}, g_{b-1}], & Z_b^{\text{align}} &= [z_b^{\text{sem}}, z_b^{\text{cam}}, z_b^{\text{temp}}]. \end{aligned} \quad (3)$$

Here α_b adaptively balances current semantic evidence and remembered context; p_b and p_{b-1} are camera tokens on the two sides of the boundary, with $\Delta p_b = p_b - p_{b-1}$; and $\bar{z}_{b-1}$, δ_{b-1} , and g_{b-1} denote the preceding alignment summary, camera-latent residual, and residual gate. These are not independent reconstruction modules. They organize heterogeneous boundary evidence so that its source remains identifiable when entering the decoder.

After adding type embeddings, Z_b^{align} is inserted after the native camera token and decoded jointly with the image, camera, and human representations by the frozen recurrent backbone. The decoded boundary representations produce gated camera and human latent residuals. The former updates the camera representation before the original pose head predicts the temporary camera $\tilde{C}_b$; the latter incorporates person-translation constraints into the same boundary inference. All typed representations and latent updates exist only in the alignment path and are never written into the state propagated through the new shot. Pooling and residual details are given in Appendix C.

We train the alignment representations with both transition and continuous four-frame sequences. Transition samples follow $A(t), A(t+1), B(t+2), B(t+3)$ for two cameras A and B , supervising changes across viewpoint transitions; continuous samples contain four successive frames from one camera and preserve the original behavior under continuous input.

The alignment objective is

$$\mathcal{L}_{\text{align}} = \mathcal{L}_t + 5\mathcal{L}_R + 10\mathcal{L}_h + 0.05(\mathcal{L}_g + \mathcal{L}_{\text{imp}}) + 10^{-5}(\mathcal{R}_{\text{cam}} + \mathcal{R}_{\text{human}}), \quad (4)$$

where $\mathcal{L}_t$ and $\mathcal{L}_h$ are Smooth-L1 camera and human translation losses, and $\mathcal{L}_R$ is quaternion geodesic distance. With e and e^{raw} denoting normalized corrected and unmodified camera errors, respectively, the gate and improvement terms are $\mathcal{L}_g = \text{BCE}(g, \text{clip}(e^{\text{raw}}, 0, 1))$ and $\mathcal{L}_{\text{imp}} = [e - e^{\text{raw}}]_+$. The remaining terms regularize camera and human latent updates. Because a transition changes the relation between history and the current observation rather than the within-shot reconstruction target, we optimize only the typed representations, camera and human latent-residual branches, and low-rank prediction-head adapters (Hu et al., 2022). The image encoder, recurrent backbone, base heads, and Multi-HMR component (Baradel et al., 2024) remain frozen.

3.3 CROSS-SHOT SPATIAL AND IDENTITY CONSISTENCY

The reinitialized path isolates new-shot recurrence from incompatible history, but its predictions inhabit an independently initialized local frame. An inter-shot relation is useful only if it can commit these predictions to the common world frame without reintroducing the preceding recurrent state. Shot3R therefore retains only the camera relation inferred at the boundary and applies it to the reinitialized outputs; the temporary alignment state has no path into later recurrence. Since the two paths process the same image, their relative camera pose captures the relation induced by historical conditioning. Let C_b^r be the camera-to-world pose from the reinitialized path, and let $\tilde{C}_b$ be the temporary alignment-path pose expressed in the accumulated common world frame. The initial mapping of the new-shot local frame is

$$T_b^{\text{align}} = \tilde{C}_b(C_b^r)^{-1}. \quad (5)$$

It satisfies $T_b^{\text{align}}C_b^r = \tilde{C}_b$ and therefore maps the reinitialized camera and human predictions into the existing frame. For later transitions, $\tilde{C}_b$ includes the previously accumulated inter-shot mapping. Importantly, T_b^{align} results from feature interaction between the current boundary observation and recurrent history, rather than post-hoc registration of two fully reconstructed shots.

Camera alignment alone does not determine person correspondence. After the initial spatial mapping, Shot3R associates people one-to-one using their predicted root locations, torso orientations, and root-centered joint configurations. For cue $a \in \{\text{root}, \text{torso}, \text{joint}\}$, we normalize its pairwise cost matrix and form the combined cost as

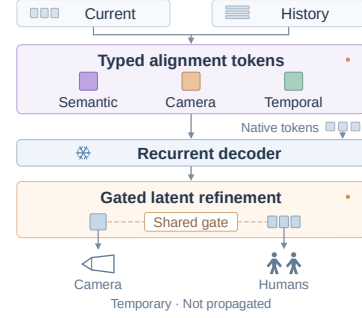


Figure 3: **History-conditioned shot alignment.** Typed representations organize current and historical evidence; gated latent updates remain temporary.

$$\hat{d}_{ij}^a = \frac{d_{ij}^a}{\max(\text{median}_{u,v} d_{uv}^a, \epsilon)}, \quad d_{ij} = \hat{d}_{ij}^{\text{root}} + \hat{d}_{ij}^{\text{torso}} + \hat{d}_{ij}^{\text{joint}}. \quad (6)$$

Here the root term is Euclidean distance, the torso term is angular distance, and the joint term averages Euclidean distances between root-centered joints. Hungarian assignment (Munkres, 1957) gives the matched set $\mathcal{M}_b$. Matching uses no identity labels, external camera calibration, or future frames. Matched people inherit their preceding anonymous identities, whereas unmatched new-shot slots receive new identities. This permutation is established once at the first new-shot frame; subsequent frames retain the native slots maintained by the reinitialized recurrence.

The initial camera mapping establishes orientation and the primary world relation, while matched people provide a direct constraint on the remaining shared translation. Let r_{b-1}^i be the predicted pelvis center of person i in the final preceding frame and $\bar{r}_b^j$ the pelvis center of person j after the initial mapping. We estimate

$$\Delta_b = \lambda \text{median}_{(i,j) \in \mathcal{M}_b} (r_{b-1}^i - \bar{r}_b^j), \quad \lambda = 0.5, \quad (7)$$

where the median is taken component-wise over matched pelvis displacements and the fixed λ is unchanged across datasets. If no valid match is available, the translation is omitted and only the camera-derived mapping is retained. With $T_b^\Delta = \begin{bmatrix} I_3 & \Delta_b \\ \mathbf{0}^\top & 1 \end{bmatrix}$ and $T_b = T_b^\Delta T_b^{\text{align}}$, the committed outputs for $t \geq b$ are

$$\begin{aligned} C_t' &= T_b C_t^r, & J_t^i &= T_b J_t^{r,i}, & V_t^i &= T_b V_t^{r,i}, \\ P_t'^w &= C_t' P_t^{r,c} = T_b P_t^{r,w}, & q_t^i &= \pi_b(q_t^{r,i}), & S_t' &= S_t^r. \end{aligned} \quad (8)$$

Here π_b is the boundary identity mapping, and 3D points are understood in homogeneous coordinates. The camera-local scene point map $P_t^{r,c}$ is left unchanged; its world-space realization follows from the transformed camera pose. Applying one shared world transform to cameras and all people preserves their relative geometry, while retaining S_t^r ensures that only the reinitialized state survives into future reconstruction. The transform is cached and applied as subsequent frames are processed. It uses no future frame, performs no sequence-level optimization, and leaves all previously emitted outputs unchanged.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Baselines. We use the publicly released Human3R checkpoint as our primary controlled baseline. Human3R and Shot3R receive the same ordered RGB frames, preprocessing, and evaluation; the principal difference is that Human3R propagates its recurrent state continuously, whereas Shot3R handles each detected transition through history-conditioned alignment and reinitialized recurrence. We additionally evaluate the official implementations of OnlineHMR, TRACE, PromptHMR, and JOSH, and include TRAM in the standardized timing study (Zhao et al., 2026; Sun et al., 2023; Wang et al., 2025c; Liu et al., 2026b; Wang et al., 2024b). Table 1 groups methods by temporal access and marks explicit transition handling; preprocessing and output scope are reported in Appendix A. HumanMM and Multi-THuMBS are included as literature context but not in the quantitative comparison because their complete code, model weights, and evaluation configurations are not publicly available.

Datasets and metrics. We construct sequential multi-shot inputs from EgoBody, EgoHumans, and Harmony4D (Zhang et al., 2022; Khiredkar et al., 2023; 2024). Their synchronized recordings are converted into monocular shots presented successively rather than simultaneously, yielding 129, 90, and 88 fixed evaluation cases, respectively. EgoBody supports controlled analysis of the boundary mechanism, EgoHumans contains substantial viewpoint changes, and Harmony4D provides crowded multi-person interactions and repeated-transition evaluations. W-MPJPE (W) fixes a Sim(3) alignment

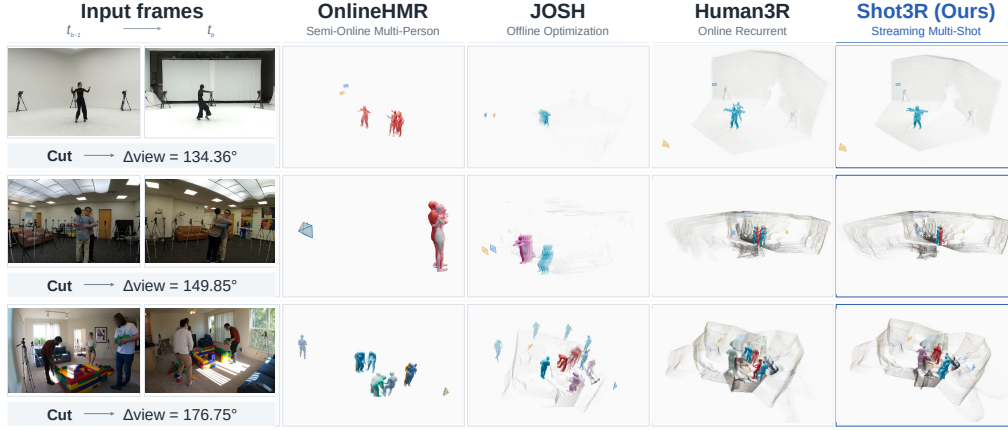


Figure 4: **Key qualitative comparisons across shot transitions.** Inputs before and after each transition are followed by OnlineHMR, JOSH, Human3R, and Shot3R. Viewpoint changes are 134.36°, 149.85°, and 176.75° from top to bottom. Shot3R maintains a more coherent camera-human layout; scene geometry is shown where available. The complete seven-method view is in Appendix D.7.

Table 1: Comparison on the EgoBody and EgoHumans multi-shot multi-person reconstruction protocols.

Category	Method	Transition	EgoBody ($N = 129$)					EgoHumans ($N = 90$)				
			W ↓	WA ↓	ATE ↓	IDF1 ↑	Cov. ↑	W ↓	WA ↓	ATE ↓	IDF1 ↑	Cov. ↑
Offline	PromptHMR (SPEC)	×	741.4 ⁶⁴	336.1 ⁸⁷	—	0.195	0.202	1193.3 ¹⁷	487.7 ²⁵	—	0.061	0.062
	JOSH	×	399.6 ¹²⁷	322.9 ¹²⁷	0.087 ¹²⁷	0.973	0.980	1410.7 ⁷⁷	481.3 ⁷⁷	1.755 ⁸⁷	0.616	0.686
Semi-online	OnlineHMR	×	961.2 ⁹⁷	418.1 ⁹⁷	0.259 ¹²⁹	0.375	0.424	1311.7 ⁵⁰	405.5 ⁵⁰	3.869 ⁸⁷	0.142	0.211
Online	TRACE	×	576.4 ²²	276.4 ³⁵	—	0.036	0.035	2464.6 ⁴³	749.5 ⁵⁹	—	0.070	0.072
	Human3R	×	321.1	233.4	0.124	0.870	0.981	1005.1	365.7	1.673	0.471	0.639
	Shot3R	✓	258.8	201.7	0.016	0.985	0.981	866.8	336.9	1.423	0.571	0.616

Transition denotes explicit shot-transition handling. W/WA are in mm and ATE in m (Sim(3) on EgoBody; SE(3) on EgoHumans). Superscripts give the number of evaluable streams for conditional geometry metrics; IDF1 and Coverage use the complete predefined protocol. Geometry means computed on different supports do not constitute a common ranking.

using the earliest valid observations and is therefore sensitive to cross-shot displacement, whereas WA-MPJPE (WA) aligns all valid observations. We further report camera trajectory error, IDF1 for cross-shot identity continuity (Ristani et al., 2016), and Coverage to account for missing predictions. Metric definitions and aggregation protocols are detailed in Appendix A.

Implementation details. Shot3R is trained exclusively on AvatarReX and THuman, which are disjoint from all evaluation benchmarks. We selectively optimize 19.62M parameters, corresponding to 1.63% of the complete 1.208B-parameter model. These trainable parameters comprise the shot-boundary alignment components and low-rank adapters, while the original pretrained backbone remains frozen. This confines adaptation to the interaction between current observations and recurrent history at shot boundaries while preserving the pretrained within-shot reconstruction prior. All benchmarks use a single checkpoint and inference configuration, without dataset- or video-specific optimization. Appendix B provides the training-data composition, optimization schedule, parameter breakdown, hardware configuration, and convergence curves.

4.2 MULTI-SHOT MULTI-PERSON RECONSTRUCTION

Table 1 reports EgoBody and EgoHumans, with the complete Harmony4D comparison in Appendix D. In the controlled comparison with Human3R, Shot3R improves WA, camera error, and IDF1 on all three datasets; W improves on EgoBody and EgoHumans and increases modestly on Harmony4D. On EgoBody, W/WA decreases from 321.1/233.4 to 258.8/201.7 mm, ATE from 0.124 to 0.016 m, and

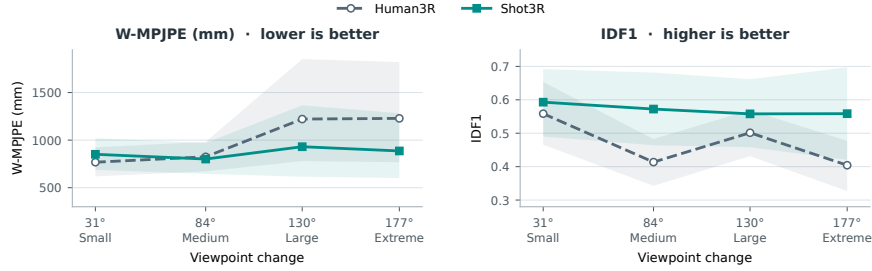


Figure 5: **Viewpoint response on EgoHumans.** Curves report absolute W-MPJPE and IDF1 in four fixed angle strata; bands are capture-bootstrap 95% confidence intervals. Shot3R improves IDF1 throughout, with the clearest W-MPJPE reductions at 130° and 177°.

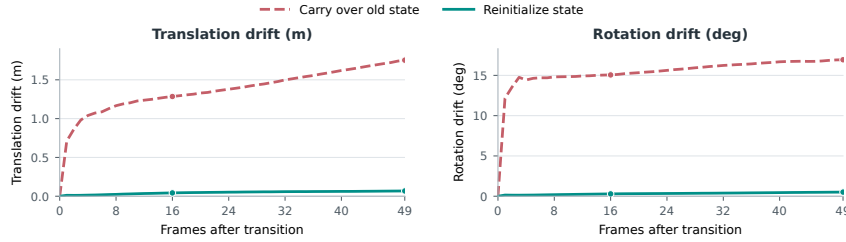


Figure 6: **Old-state carry-over causes within-shot camera drift.** On static-camera captures, the same unmodified Human3R checkpoint processes identical new-shot RGB with no Shot3R component. First-prediction anchoring removes any fixed transform; carry-over then drifts while reinitialization remains stable.

IDF1 rises from 0.870 to 0.985 at unchanged 0.981 Coverage. Local pose and mesh accuracy remain essentially unchanged (Appendix Table 26), indicating that the gains arise primarily from spatial placement in the common frame, camera estimation, and identity continuity rather than relearning within-shot body reconstruction. The public-system comparison provides complementary evidence: on shared valid inputs, Shot3R lowers human and camera errors relative to OnlineHMR and raises IDF1 and Coverage over complete streams; it also yields lower W/WA/ATE than the offline JOSH pipeline on cases evaluable by both methods across all three datasets, without future-frame access. Because TRACE and PromptHMR return evaluable geometry for only subsets of EgoBody, their conditional errors must be interpreted together with valid support; complete paired comparisons and failure records are provided in the appendix.

The advantage becomes more pronounced as the inter-shot viewpoint change increases. With equal weight per dataset, Shot3R reduces W-MPJPE by 9.5% over all viewpoints and by 19.1% on the predefined large-viewpoint subsets. On extreme-view EgoBody, W-MPJPE decreases by 29.9%, while IDF1 rises from 0.758 to 0.983. Figure 4 further shows transitions of 134.36°, 149.85°, and 176.75°: despite pronounced viewpoint and appearance changes, Shot3R retains a more coherent camera-human layout. Figure 5 and the appendix report the complete viewpoint strata and continuous-angle analysis.

4.3 ANALYSIS AND ABLATION STUDIES

We first isolate the effect of state carry-over using the same original Human3R checkpoint on identical post-transition RGB in 90 EgoHumans cases captured by a static physical camera, without any Shot3R component. Both camera trajectories are anchored to their first new-shot prediction, removing any fixed coordinate transform. Over the next 16 frames, carrying the old state produces 1.124 m/14.58° translation/rotation drift, compared with 0.026 m/0.20° after reinitialization; the latter is lower on all 27 captures. Old-state carry-over therefore induces time-varying within-shot drift, not merely a fixed coordinate offset. Confidence intervals, window-length controls, and paired tests are reported in Appendix D.5.1.

Table 2: Alignment–state-propagation 2×2 control and the complete Shot3R on EgoBody.

Configuration	Boundary cue	Shot-transition operation			EgoBody metrics			
		Alignment	Propagated state	Association	W (mm) ↓	WA (mm) ↓	ATE (m) ↓	IDF1 ↑
Human3R	—	×	Continued	×	321.1	233.4	0.124	0.870
State reinitialization	Annotated	×	Reinitialized	×	752.6	589.4	1.141	0.834
Alignment + continued state	Annotated	✓	Continued	×	293.3	227.3	0.082	0.868
Alignment + reinitialized state	Annotated	✓	Reinitialized	×	352.8	267.7	0.017	0.834
Shot3R	Detected	✓	Reinitialized	✓	258.8	201.7	0.016	0.985

Annotated boundaries are used only by the controlled configurations; Human3R receives no boundary cue. The two aligned controls disable association and person-guided translation, so only the state propagated after alignment changes.

Table 3: Geometric registration controls on the complete EgoBody and EgoHumans test sets. W is in mm and ATE in m. Fit denotes the success rate of the separate geometric fit on EgoHumans; Shot3R has no such stage.

Method	Access	EgoBody ($N = 129$)			EgoHumans ($N = 90$)			
		W ↓	ATE ↓	IDF1 ↑	W ↓	ATE ↓	IDF1 ↑	Fit ↑
Human3R per-shot reset	Independent shots	752.6	1.141	0.834	1248.1	4.411	0.455	—
+ pelvis translation	Boundary-time	500.2	0.068	0.896	1242.1	3.326	0.506	100.0%
+ human-joint SE(3)	Boundary-time	261.5	0.019	0.896	897.0	1.731	0.510	82.2%
+ scene FPFH+ICP	Offline full shots	733.2	0.020	0.834	1372.8	3.847	0.455	41.1%
Shot3R	Streaming	258.8	0.016	0.985	866.8	1.423	0.571	—

Traditional controls receive annotated shot boundaries and never use ground truth to estimate a transform. Failed fits fall back to the per-shot-reset result and remain in the fixed denominator. EgoBody fitting succeeds for every case.

Table 2 separates the resulting failure modes. Reinitialization alone loses the inter-shot spatial relation, yielding 752.6/589.4 mm W/WA and 1.141 m ATE. Once alignment is introduced, continued state gives lower W/WA than reinitialized state (293.3/227.3 versus 352.8/267.7 mm), whereas reinitialization reduces ATE from 0.082 to 0.017 m. Alignment thus restores the relation between shots, while reinitialization stabilizes recurrence within the new shot; neither choice alone resolves both failure modes. The complete Shot3R additionally uses person association to support identity continuity and a shared camera–human transform, reaching 258.8/201.7 mm W/WA, 0.016 m ATE, and 0.985 IDF1. These values reflect the combined design rather than the effect of any single component, and support treating alignment and recurrent-state propagation as separate decisions.

Geometric registration controls. We next test whether explicit geometric fitting after independent per-shot reconstruction can recover the same cross-shot result. Starting from identical per-shot-reset predictions, we fit a shared pelvis translation, a robust human-joint SE(3), or an offline FPFH/RANSAC+ICP transform from the complete shot point clouds. All three controls receive annotated transitions and use predictions only; Shot3R retains its online detector. As Table 3 shows, Shot3R obtains the lowest full-set W and ATE and the highest IDF1 on both datasets. Human-joint SE(3) is the strongest geometric control and approaches Shot3R in W, but has lower identity consistency and fails to fit 17.8% of EgoHumans cases. Thus a strong human-guided fit can recover much of the spatial relation, whereas Shot3R provides the more balanced human, camera, and identity result without a separate fitting stage. Full WA/Coverage results, paired intervals, fixed viewpoint strata, and fitting diagnostics are reported in Appendix D.3.

Supporting analyses examine association, repeated transitions, and efficiency. On Harmony4D, augmenting pelvis position with torso orientation and root-centered joints raises identity continuation from 78.41% to 80.68% and IDF1 from 0.620 to 0.635, indicating that these predicted geometric cues are complementary. A small three-shot control shows that the same boundary operation can be reapplied, without establishing drift-free behavior over long sequences (Appendix D.6.3). The online detector identifies the annotated boundary in every primary-protocol stream. Under standardized reconstruction timing on one NVIDIA L20, Shot3R and Human3R achieve comparable throughput at 3.106 and 3.224 FPS, respectively. The complete association ablation, transition-count analysis, and seven-method full-system timing are provided in Appendix B.

5 CONCLUSION

We study streaming reconstruction across shot transitions, where recurrent history plays conflicting roles: it provides the reference needed to relate successive shots, yet its continued propagation can introduce time-varying drift within the new shot. Our key insight is that history should be temporarily read for inter-shot alignment without being propagated as the recurrent state of the new shot. Based on this insight, Shot3R infers the inter-shot relation from pre-transition history while allowing only an independently reinitialized state to drive subsequent reconstruction. It expresses the cameras, humans, and scene geometry from each shot in the common world frame and maintains person identities while processing frames in arrival order, without future-frame access or per-video optimization. Experiments on three multi-person benchmarks demonstrate consistent improvements in world-frame human reconstruction, camera trajectory estimation, and identity association, with the largest gains under substantial viewpoint changes. Handling long videos with complex edits, repeated entrances and exits, and rapid visual changes remains an important direction for future work.

AI USE STATEMENT

Language-model assistance was used to search and summarize relevant literature; plan and edit the manuscript structure and prose; translate between Chinese and English; discuss experimental design and interpret diagnostic results; scaffold and review analysis and document-tooling code; and prepare editable vector layouts and annotations. The authors independently verified cited sources, code, protocols, numerical results, figures, and claims against the underlying records. Generative AI was not used to produce ground truth, evaluation inputs, or reconstruction results. The authors take responsibility for the final content of this work.

ETHICS STATEMENT

The human-capture datasets are public or research-licensed and are used under their providers' consent, access, and research-use conditions. We do not infer participant identity, link participants across datasets, or use the material for surveillance. Restricted imagery, annotations, and checkpoints remain in access-controlled storage and are not redistributed when prohibited by their licences.

REPRODUCIBILITY STATEMENT

Appendix A specifies the datasets, inputs, metrics, and comparison protocol; Appendix B reports training and implementation details; and Appendices C and D provide the method details, controlled analyses, and complete results. Tables are generated from machine-readable summaries with source hashes. End-to-end reproduction additionally requires the licensed data and checkpoints. Release materials will include code, protocol manifests, aggregate measurements, and non-identifying case identifiers to the extent permitted by those licences.

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Shot3R: Multi-Shot State Transitions for Streaming 4D Human-Scene Reconstruction

Supplementary Material

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SUPPLEMENTARY CONTENTS

The supplement is organized into three parts. Part I defines the evaluation contract and documents training, inference, detector, runtime, and coordinate conventions. Part II provides the complete comparisons and controlled evidence under that contract. Part III states the supported scope and limitations. Page numbers link to the corresponding sections.

Part I: Protocols, Method Details, and Inference	18
A Evaluation Protocols	18
B Training and Implementation Details	20
C Additional Method Details	22
Part II: Additional Experimental Evidence	25
D Additional Experiments	25
D.1 Large Viewpoint Changes	25
D.2 Comparison with Public Methods	27
D.3 Traditional Geometric Registration Controls	30
D.4 EgoBody	34
D.5 EgoHumans	36
D.5.1 Recurrent-State Carry-Over	36
D.6 Harmony4D	38
D.6.1 Shared-Translation Weight	38
D.6.2 Cross-Shot Person Association	38
D.6.3 Three-Shot Control	38
D.7 Full Qualitative Comparison	40
Part III: Scope and Limitations	41
E Limitations	41

Highlights and suggested reading. For a compact path through the principal evidence, we recommend:

Method and reproducibility. Section A fixes datasets, metrics, and comparison scope; Sections B–C document trainable parameters, detector, runtime, coordinate transforms, and repeated inference.

State-transition evidence. Section D.5.1 gives the same-checkpoint carry-over control; Sections D.4–D.6 provide the corresponding reconstruction and association analyses.

Viewpoint and repeated-transition evidence. Section D.1 analyzes viewpoint change, and Section D.6.3 verifies that the transition operation can be reapplied on three-shot streams.

Geometric registration controls. Section D.3 compares independent reconstruction followed by translation, human-joint SE(3), and full-shot scene registration.

Visual and scope evidence. Section D.7 retains the full seven-method qualitative comparison, while Section E states the supported scope and remaining limitations.

Part I: Protocols, Method Details, and Inference

A EVALUATION PROTOCOLS

Datasets. The three main benchmarks are constructed from synchronized multi-camera captures, but are presented to every method as temporally ordered monocular videos: only one view is observed

Table 4: Datasets used in the multi-shot multi-person evaluation.

Dataset	Videos	Captures	People	Shots	Aggregation
EgoBody	129	43	Multiple	2	Recording
EgoHumans	90	27	Multiple	2	Video
Harmony4D	88	25	Multiple	2	Video

Table 5: Temporal access, preprocessing, and native output scope of the evaluated public methods.

Access	Method	Preproc.-free	Camera	Scene	Transition
Offline	PromptHMR (SPEC)	×	×	×	×
	JOSH	×	✓	✓	×
	TRAM	×	✓	×	×
Semi-online	OnlineHMR	×	✓	×	×
Online	TRACE	✓	×	×	×
	Human3R	✓	✓	✓	×
	Shot3R	✓	✓	✓	✓

A check mark reports an available property or native output, not reconstruction quality. Preproc.-free excludes precomputed detection, tracking, camera-frame humans, camera, mask, depth, and contact cues. TRAM is evaluated only in the runtime protocol.

at any time. EgoBody, EgoHumans, and Harmony4D contain 129, 90, and 88 two-shot videos, respectively. Table 4 summarizes the three evaluation sets.

Input and supervision. All methods within a dataset receive the same RGB sequence. Camera calibration, 3D bodies, person identities, and annotated transition locations are used only for evaluation and are not given to the model. The only exceptions are the two controlled intermediate configurations in Table 2, which use annotated transition timing to examine the effects of state reinitialization and alignment under a controlled boundary setting.

Metrics. W-MPJPE (W) estimates a Sim(3) transform from the earliest valid observations and keeps it fixed for subsequent frames; WA-MPJPE (WA) estimates Sim(3) from all valid observations. We additionally report camera absolute trajectory error (ATE), IDF1 for identity consistency, and Coverage for the fraction of ground-truth person instances represented by valid predictions. Because the alignment definitions differ across datasets, absolute errors are compared only within the same dataset.

EgoBody is aggregated within each recording and then averaged over its 43 recordings. EgoHumans and Harmony4D use clip-level means. Geometry errors that require a valid person match are always accompanied by their number of valid clips or by Coverage.

Public methods and offline reference. OnlineHMR, TRACE, and PromptHMR are evaluated with their released implementations on the same RGB sequences as Shot3R. OnlineHMR provides a separate camera trajectory; the released TRACE and PromptHMR outputs are evaluated for human reconstruction under our protocol. Parenthesized counts in Table 1 show the number of clips with valid geometry. OnlineHMR combines semi-online tracking with a global camera backend, whereas PromptHMR processes the complete video; these temporal differences are not treated as equivalent to Shot3R’s streaming inference. JOSH uses its released implementation with Pi3X initialization and one fixed configuration across all three main protocols. It receives each complete RGB sequence offline, but no ground-truth camera, identity, body shape, transition, or shot-wise transform; we do not manually stitch identities. Failures remain in complete-dataset identity and Coverage denominators, and unavailable errors are not imputed. TRAM is run with its released full prediction path for the efficiency study; this profiling run is not included in the three-dataset accuracy comparison.

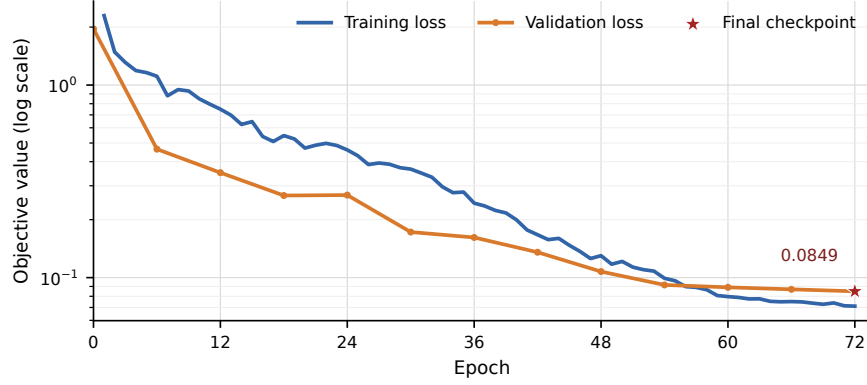


Figure 7: Training and validation objectives of the reported Shot3R model. The curves use the aggregation rules defined above; the star marks the epoch-72 checkpoint used throughout.

B TRAINING AND IMPLEMENTATION DETAILS

Training configuration. The reported Shot3R model is initialized directly from pretrained Human3R and trained on the lbn1/zzr subjects of AvatarReX and THuman00/02. Each source contains 8,000 shot-transition clips and 2,000 continuous clips. A shot-transition clip follows $A(t), A(t+1), B(t+2), B(t+3)$, whereas a continuous clip contains four consecutive frames from one camera; both carry camera and human annotations in a shared world frame.

The image encoder, recurrent decoder, point-map head, base camera and human heads, and Multi-HMR backbone (Baradel et al., 2024) are frozen. The trainable shot-alignment module comprises a 14.39M alignment block with three typed tokens, a 1.78M camera latent-residual head, a 2.96M human latent-residual head, and 0.50M rank-8 LoRA parameters in the camera and human heads (Hu et al., 2022): 19.62M parameters in total, or 1.63% of the 1.208B-parameter model. The three alignment tokens occupy the 768-dimensional decoder space, and the alignment gate is shared with the human residual branch.

Each epoch contains 100 updates with ten four-frame sequences per batch; AvatarReX and THuman are sampled with probabilities 0.6 and 0.4. Training runs for 72 epochs and 7,200 optimizer updates. AdamW uses $\beta = (0.9, 0.95)$, weight decay 0.05, four-epoch linear warm-up to 2×10^{-4} , and cosine decay to 10^{-6} , with automatic mixed precision and gradient checkpointing.

Let e and e^{raw} be the corrected and uncorrected camera errors after normalizing translation by 0.5 m and rotation by 45° . We optimize the shot-alignment module with

$$\begin{aligned} \mathcal{L}_{\text{align}} = & \mathcal{L}_t + 5\mathcal{L}_R + 0.05 \text{BCE}(g, \text{clip}(e^{\text{raw}}, 0, 1)) \\ & + 0.05[e - e^{\text{raw}}]_+ + 10^{-5}\mathcal{R}_{\text{cam}} + 10\mathcal{L}_h + 10^{-5}\mathcal{R}_{\text{human}}, \end{aligned} \quad (9)$$

where $\mathcal{L}_t$ and $\mathcal{L}_h$ are Smooth-L1 camera and human translation losses, $\mathcal{L}_R$ is quaternion geodesic distance, g is the learned alignment gate, and the residual terms constrain the camera and human latent updates. Camera and human supervision is evaluated across the four-frame training sequence; no ground truth is used at inference.

Training convergence. Validation is run every six epochs on four fixed splits defined by dataset (AvatarReX or THuman) and clip type (shot transition or continuous). Figure 7 plots the global mean training objective and the mean of the four validation median losses. The epoch-72 checkpoint is used in all reported experiments.

Shot-transition detection. The detector computes 20-dimensional colour, grayscale, hash, edge, optical-flow, histogram, and ORB statistics from consecutive RGB frames and retains the most recent three frame pairs. A single-layer GRU (Cho et al., 2014) with a 32-dimensional hidden state and a linear binary head predicts the transition probability. It is trained for 120 epochs using binary cross-entropy and AdamW with a learning rate of 2×10^{-3} and weight decay of 2×10^{-3} (seed 20260805).

Table 6: Full-system runtime on five fixed 150-frame Harmony4D multi-shot streams (750 frames total).

Category	Method	Camera	Scene	s / 150 frames ↓	FPS ↑
Offline	PromptHMR (SPEC)	×	×	708.7 ± 209.0	0.212
	TRAM	✓	×	455.5 ± 60.6	0.329
	JOSH	✓	✓	949.6 ± 286.0	0.158
Semi-online	OnlineHMR	✓	×	1007.8 ± 124.7	0.149
Online	TRACE	×	×	45.0 ± 21.5	3.334
	Human3R	✓	✓	120.4 ± 3.2	1.246
	Shot3R	✓	✓	167.8 ± 4.0	0.894

Seconds are mean ± population standard deviation; FPS is 750 divided by total wall time. Camera denotes an independently evaluable physical-camera trajectory, and Scene denotes scene geometry. Timing includes required prediction-process launches and model loading through native prediction save; multi-stage releases report the sum of their required stages. Rendering and evaluation are excluded. Output scope therefore remains relevant when interpreting throughput.

Table 7: Reconstruction time and peak memory for one 100-frame EgoHumans clip on an NVIDIA L20 (resolution 512, batch size 1, FP32).

Method	s / 100 frames ↓	FPS ↑	Peak VRAM (GiB) ↓
Human3R	31.020	3.224	5.292
Shot3R, no transition	31.125	3.213	5.359
Shot3R, one transition	32.191	3.106	5.246

The decision threshold is 0.5. The frozen training ledger contains 408 frame pairs (216 positive) from eight constructed patterns spanning AvatarReX, THuman, and two MVHuman transition settings. In the formal two-shot protocol, the first score above threshold triggers the state change. No future frames, camera calibration, body annotations, or person identities enter the detector.

The first trigger coincides with the annotated boundary in all 129 EgoBody, 90 EgoHumans, and 88 Harmony4D streams. Harmony4D has no off-boundary positive response across 13,200 scored frames.

Full-system runtime. Table 6 reports full-system wall time for seven methods on five fixed 150-frame Harmony4D multi-shot streams. The set was frozen before runtime aggregation and contains hugging, grappling2, sword2, mma4, and mma5; each input comprises two sequential 75-frame monocular shots. Every required prediction stage uses one NVIDIA L20, and timing includes its fresh process launch and model loading through native prediction serialization. For releases with multiple required stages, we sum their stage times.

The scope includes model construction and checkpoint loading, plus any detection, segmentation, tracking, camera recovery, temporal/global optimization, and human recovery required by the released prediction path. The retained PromptHMR and TRACE runs also include lightweight per-case hardlink staging, without image decoding or re-encoding. Dataset download and archive extraction, metric evaluation, paper-format conversion, and rendering are excluded. FPS is the micro-average $750 / \sum_i t_i$; seconds report the mean and population standard deviation across the five clips. All 35 method-clip runs completed successfully; a failed run would not have been silently omitted.

Controlled reconstruction cost. Table 7 separately measures the controlled reconstruction component on one 100-frame EgoHumans clip. Timing includes model inference, output decoding, person association, and world-coordinate alignment, but excludes image loading, resizing, and transition detection. It isolates the cost of the transition operation and is not mixed with the full-system FPS in Table 6. A single transition adds 1.066 s (3.42%) relative to Shot3R without a transition.

Transition-count control. To isolate the number of shot boundaries, we duplicate one fixed 150-frame EgoBody RGB input to form the same 300-output-frame timeline for every configuration and place 0, 1, 3, or 5 annotated transitions. Each boundary requires one additional evaluation of its

Table 8: Controlled Shot3R runtime and peak allocation versus the number of transitions, with 300 output frames. Peak is the maximum PyTorch allocated memory over the three measured runs.

Transitions	Neural evaluations	Median time (s)	Output FPS	Peak (GiB)
0	300	105.423	2.846	5.82
1	301	107.286	2.796	5.48
3	303	103.466	2.900	5.30
5	305	102.585	2.924	5.25

boundary frame, so C transitions execute exactly $300 + C$ neural frames. Hardware, precision, and timing scope match the controlled reconstruction test above; each configuration has one warm-up followed by three measured repetitions, from which we report the median.

Median time spans 102.6–107.3 s and does not increase monotonically with the number of transitions; peak allocation does not increase either. Thus, within this test range, up to five boundary operations do not produce a throughput change distinguishable from the measured execution variation. Because more transitions also split inference into shorter segments, the lower peaks must not be interpreted as constant full-system memory for arbitrarily long sequences.

C ADDITIONAL METHOD DETAILS

Shot-alignment module. At the first frame of a new shot, the history-conditioned alignment path inserts three tokens after the backbone camera token. A learned query attends to the projected current image tokens, the camera token, and available human tokens to obtain c_b . A second query attention pools the recurrent state tokens, camera memory, and preceding decoder-refined alignment tokens into m_{b-1} . The semantic mixing weight in Eq. 3 is $\alpha_b = \sigma(\text{MLP}_{\text{sem}}[c_b, m_{b-1}])$.

Here $\Delta p_b = p_b - p_{b-1}$; $\bar{z}_{b-1}$, δ_{b-1} , and g_{b-1} in Eq. 3 are the preceding alignment summary, camera correction, and gate.

The camera-alignment token is produced from the current camera token, the preceding one, their latent difference, and m_{b-1} . The temporal-context token pools the preceding decoder-refined alignment tokens, camera residual, and residual gate. A learned type embedding distinguishes the three roles; LayerNorm is applied before decoder attention refines them jointly with the backbone image, camera, and human tokens.

A learned residual updates the temporary camera representation before it is decoded by the frozen camera head. The human residual branch supplies additional supervision during training. Once the inter-shot transform has been estimated, the temporary predictions and state are discarded; only the reinitialized state is propagated through the new shot.

Person association. For a pre-transition person i and post-transition person j , the matching cost combines root position, torso orientation, and root-centered joint configuration:

$$d_{ij} = \hat{d}_{\text{root}}(i, j) + \hat{d}_{\text{torso}}(i, j) + \hat{d}_{\text{joint}}(i, j). \quad (10)$$

Each term is normalized by the median of its valid pairwise values. Hungarian assignment returns $\min(N^-, N^+)$ one-to-one pairs, and no confidence threshold is applied. Later frames retain the identities initialized at the first post-transition frame. The matching uses predictions only, without ground-truth identity, calibration, or future frames.

Shared camera-human transform. The inter-shot camera transform first maps the new shot into the existing world frame. The median displacement of matched people then provides the shared translation in Eq. 7. The resulting transform is applied equally to camera poses, joints, and mesh vertices. Unmatched new-shot detections receive new anonymous identities. If no pair is available, the shared translation is omitted and only camera alignment is retained.

Coordinate convention. All camera poses are homogeneous camera-to-world transforms $C = \begin{bmatrix} R & t \\ \mathbf{0}^\top & 1 \end{bmatrix}$. A world transform T is left-composed with the camera, world-space scene points, and

human geometry: $C \leftarrow TC$, $P^w \leftarrow TP^w$, and $X \leftarrow TX$, with all points expressed in homogeneous coordinates. The alignment module does not deform the camera-local scene geometry P^c ; its world-space realization is obtained from the corrected camera pose, $P^w = CP^c$, and is therefore transformed consistently with the camera and human predictions.

Multiple transitions. At every later transition, the current recurrent state is used only to estimate the new spatial relation, while a newly initialized state supplies subsequent recurrence. Previously emitted frames remain unchanged. For $b_1 < b_2$, the first transform affects frames from b_1 onward, and the second affects only frames from b_2 onward; later predictions use the composed transform T_2T_1 . Section D.6.3 evaluates this composition on three-shot sequences.

Part II: Additional Experimental Evidence

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D ADDITIONAL EXPERIMENTS

This section follows the main questions of the paper: performance under large viewpoint changes, comparison with public methods and traditional geometric registration, repeated shot transitions, and detailed results on the three multi-person benchmarks. Tables already reported in the main paper are not repeated unless additional metrics are required for a complete protocol audit.

D.1 LARGE VIEWPOINT CHANGES

The main paper reports the dataset-balanced summary. Table 9 further gives the results on the predefined farthest-view subset of each dataset. The subsets contain 43 EgoBody recordings, 22 EgoHumans captures, and 25 Harmony4D captures and are defined without reference to method performance.

To test whether the advantage is more pronounced at large viewpoint changes, we compare the paired gain in the large-viewpoint subset with the gain on the remaining views. Positive values in Table 10 indicate a larger Shot3R gain under larger viewpoint changes. The dataset-balanced intervals are positive for W, WA, and IDF1, while individual datasets need not improve on every metric.

To examine the camera-level relation directly, we use the exact camera angle for all 90 EgoHumans cases and correlate viewpoint change with the Shot3R gain over Human3R. Intervals use a two-stage bootstrap that first resamples the 27 capture clusters and then cases within each selected capture (Davison & Hinkley, 1997).

Continuous-angle analysis supports a stronger relationship for camera and WA errors, consistent with Shot3R acting on the spatial relation at a shot transition, but does not establish monotonic improvement for every human or identity metric. The stratified results retain the degradations observed in the small and medium ranges.

Table 9: Results on the predefined large-viewpoint subsets. W/WA are in mm and camera ATE is in m (Sim(3) for EgoBody/Harmony4D and SE(3) for EgoHumans).

Dataset	Method	W↓	WA↓	Camera↓	IDF1↑	Cov.↑
EgoBody	Human3R	505.1	321.8	0.253	0.758	0.974
	Shot3R	354.3 (29.9%↓)	273.3 (15.1%↓)	0.016	0.983	0.974
EgoHumans	Human3R	1228.4	400.0	3.390	0.404	0.611
	Shot3R	886.1 (27.9%↓)	315.9 (21.0%↓)	2.326	0.559	0.576
Harmony4D	Human3R	550.2	325.7	0.250	0.418	0.866
	Shot3R	552.6 (0.4%↑)	260.7 (20.0%↓)	0.017	0.622	0.859

Table 10: Additional gain at large viewpoints relative to the remaining views, with 95% recording/capture-cluster bootstrap intervals.

Dataset	W (mm)	WA (mm)	IDF1
EgoBody	+132.8 [+10.3, +286.1]	+25.2 [−12.9, +68.6]	+0.166 [+0.109, +0.224]
EgoHumans	+270.1 [−13.7, +611.8]	+73.2 [+7.0, +144.9]	+0.072 [−0.010, +0.160]
Harmony4D	+28.7 [−39.3, +102.4]	+73.2 [+35.1, +114.5]	+0.169 [+0.083, +0.253]
Dataset-balanced	+143.9 [+36.1, +269.5]	+57.2 [+28.6, +87.5]	+0.136 [+0.092, +0.180]

Table 11: Relation between viewpoint change and reconstruction gain on EgoHumans. Left: case-level Spearman correlation with two-stage capture-cluster bootstrap intervals. Right: mean gains in the predefined angle strata. W/WA are in mm, camera rotation in degrees, and camera translation in m; positive values favour Shot3R.

(a) Continuous-angle correlation

Gain quantity	ρ	95% CI
W-MPJPE	0.266	[−0.011, 0.510]
WA-MPJPE	0.287	[0.021, 0.519]
Boundary camera rotation	0.413	[0.126, 0.668]
Boundary camera translation	0.519	[0.300, 0.684]
IDF1	0.104	[−0.174, 0.368]

(b) Mean gain by predefined angle stratum

Stratum	W	WA	Cam. R	Cam. T	IDF1
Small	−83.1	−25.7	−0.11	−0.306	0.034
Medium	23.5	4.4	−1.02	−0.984	0.159
Large	290.5	57.5	10.93	0.832	0.057
Extreme	342.4	84.1	37.26	2.542	0.154

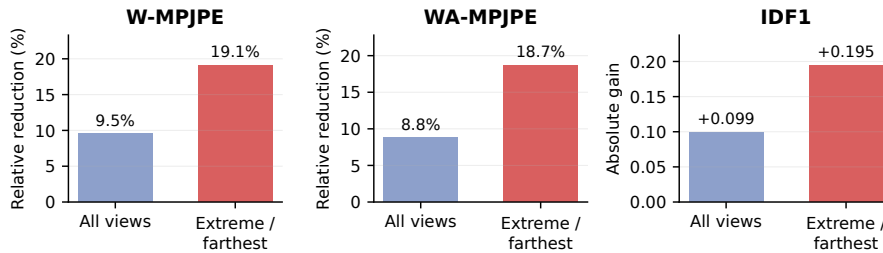


Figure 8: Dataset-balanced improvements over Human3R on all views and on the predefined large-viewpoint subsets. W/WA show relative reductions; IDF1 shows absolute gain.

D.2 COMPARISON WITH PUBLIC METHODS

Table 1 reports every executed method on the same EgoBody and EgoHumans RGB sequences, while Table 12 gives the complete Harmony4D comparison. We additionally compare Shot3R and OnlineHMR on the clips for which OnlineHMR provides valid geometry; IDF1 and Coverage are computed over all sequences. Table 15 shows lower human and camera errors for Shot3R on all three datasets. All intervals exclude zero except WA on EgoHumans. OnlineHMR uses a semi-online tracker and a separate global-camera backend, so this comparison does not imply identical temporal access.

Beyond the three primary protocols, four Harmony4D three-shot streams compare repeated transition handling in Shot3R, Human3R, and OnlineHMR; Section D.6.3 reports their complete results.

The official JOSH release is run on all 307 streams from the three primary protocols. Inference and coordinate checks complete for 302 streams; W/WA are evaluable on 292 and camera ATE on 302. Two EgoBody streams fail because contact-point initialization returns an empty set, and three EgoHumans streams fail because a shot-boundary resolution change is incompatible with the released SAM3 mask pipeline; all 88 Harmony4D streams succeed. Ten otherwise completed EgoHumans streams lack the initial human match required by the fixed W/WA evaluation protocol. Table 13 distinguishes failed runs from completed runs that cannot be evaluated by a particular metric.

To remove ambiguity from conditional means computed on different supports, Table 14 restricts each metric to streams evaluable by both Shot3R and JOSH. Shot3R obtains lower W, WA, and camera ATE on all three common-support sets while processing frames without future access. IDF1 and Coverage retain the complete protocol denominator: JOSH is higher on both in EgoHumans, whereas Harmony4D combines higher Shot3R IDF1 with higher JOSH Coverage.

For some of the closest published methods, the released resources are insufficient to evaluate our fixed two-shot sequences under the same protocol. Table 16 therefore distinguishes task scope, temporal access, and whether a release provides the outputs required by our evaluation.

Table 12: Complete comparison on the Harmony4D multi-shot multi-person protocol ($N = 88$).

Category	Method	W ↓	WA ↓	ATE-Sim3 ↓	IDF1 ↑	Coverage ↑
Offline	PromptHMR (SPEC)	963.2 (5)	306.9 (26)	–	0.030	0.034
	JOSH	770.2 (88)	369.7 (88)	0.153 (88)	0.551	0.959
Semi-online	OnlineHMR	750.8 (30)	335.9 (30)	1.364 (88)	0.068	0.159
Online	TRACE	677.1 (4)	270.2 (26)	–	0.016	0.016
	Human3R	496.6	259.6	0.1420	0.553	<u>0.915</u>
	Shot3R	519.6	247.0	0.0168	0.636	0.913

W/WA are in mm and ATE is in m. Parentheses give the number of streams evaluable for each conditional geometry metric; IDF1 and Coverage use all 88 fixed streams. Geometry means with different support do not form a common ranking.

Table 13: JOSH execution and evaluable cases on the three primary protocols. Failed runs remain in the IDF1/Coverage calculations over all sequences; unavailable human or camera errors are omitted rather than set to zero.

Dataset	Total	Completed	W/WA evaluable	ATE evaluable	Failed
EgoBody	129	127	127	127	2 contact initialization
EgoHumans	90	87	77	87	3 mask-size mismatch [†]
Harmony4D	88	88	88	88	None
Total	307	302	292	302	5

[†]Ten additional completed streams lack the initial match required by W/WA.

Table 14: Case-macro Shot3R–JOSH comparison restricted to streams evaluable by both methods. W/WA are in mm; ATE is Sim(3)-aligned for EgoBody and Harmony4D and SE(3)-aligned for EgoHumans, in m.

Dataset	W-MPJPE			WA-MPJPE		ATE		
	N	Shot3R	JOSH	Shot3R	JOSH	N	Shot3R	JOSH
EgoBody	127	260.2	402.0	202.4	324.9	127	0.017	0.088
EgoHumans	77	887.2	1410.7	337.7	481.3	87	1.289	1.755
Harmony4D	88	519.6	770.2	247.0	369.7	88	0.017	0.154

Table 15: Paired Shot3R gain over OnlineHMR with 95% clustered bootstrap intervals. Positive values favour Shot3R.

Dataset	W gain [95% CI]	WA gain [95% CI]	ATE* gain [95% CI]
EgoBody	730.1 [522.9, 941.6]	233.9 [158.8, 307.9]	0.243 [0.153, 0.348]
EgoHumans	430.3 [128.9, 714.2]	71.5 [-1.3, 139.5]	2.447 [1.906, 3.080]
Harmony4D	340.8 [165.9, 509.2]	134.0 [78.5, 190.3]	1.348 [1.227, 1.463]

Dataset	IDF1 gain [95% CI]	Coverage gain [95% CI]
EgoBody	0.610 [0.527, 0.691]	0.557 [0.465, 0.647]
EgoHumans	0.429 [0.369, 0.486]	0.405 [0.332, 0.474]
Harmony4D	0.568 [0.491, 0.640]	0.754 [0.654, 0.835]

Table 16: Task scope and released-resource compatibility for related methods not included in our numerical comparison on identical sequences. ✓ denotes an available capability, ✗ denotes an unavailable capability, and △ denotes a released resource not yet validated under our protocol; † denotes a capability reported in the original paper but not evaluable under our protocol.

Work	Shot trans.	Multi-person	Phys. cam.	Shared world	Eval. here	Input / access	Role here
HumanMM (Zhang et al., 2025b)	✓	✗	✓ [†]	✓ [†]	✗	Full sequence	Related work
Multi-THuMBS (On et al., 2026)	✓	✓	✓ [†]	✓ [†]	✗	Offline refinement	Related work
Multishot (Pavlakos et al., 2022b)	✓	✗	✗	✗	✗	Track-conditioned offline	Related work
SLAHMR (Ye et al., 2023)	✗	✓	✓	✗	✗	Single-shot offline	Different setting
WHAC / GVHMR (Yin et al., 2024; Shen et al., 2024)	✗	✗	✓	✗	✗	Full-video inference	Single-person setting
Yang et al. (Yang et al., 2026)	✗	✓	✓	✓	✗	Joint multi-view set	Multi-view setting

Evaluation under our protocol requires human tracks, a camera trajectory, identity assignments, and a common timeline spanning both shots, without ground-truth track selection, manual identity stitching, or a separately fitted post-transition transform. SLAHMR’s released configuration processes one selected shot; the released Multishot demo requires a specified person track. Physical-camera support for SLAHMR is confined to its selected shot. JOSH is no longer listed here because its released full-video implementation is evaluated in Table 1.

D.3 TRADITIONAL GEOMETRIC REGISTRATION CONTROLS

We test whether explicit geometric fitting between independently reconstructed shots can substitute for the online state transition in Shot3R. The R0 control reconstructs both shots independently with the public Human3R checkpoint. R1 matches predicted boundary people and estimates one coordinate-wise median pelvis translation. R2 uses the same prediction-only matching, followed by RANSAC and weighted Kabsch fitting over stable predicted body joints (Fischler & Bolles, 1981; Kabsch, 1976). R3 performs offline FPFH mutual matching and global RANSAC on the two complete shot point clouds, followed by point-to-plane ICP (Rusu et al., 2009; Besl & McKay, 1992). In each case, one shared transform is applied to all cameras, people, meshes, and scene points after the transition.

All geometric controls receive annotated shot boundaries, whereas Shot3R uses its online detector. A single dataset-level setting for each control is selected on the complete Development split, passed once through a disjoint Holdout gate, and then frozen. Test data are never used for tuning, ground-truth registration, or per-case method selection. Every failed fit is an exact R0 fallback and remains in the fixed denominator. Table 17 reports the complete aggregate results.

Pelvis translation is insufficient under large viewpoint rotation. Scene registration can yield a low camera ATE without placing people correctly: on EgoBody it reaches 0.020,m ATE but retains 733.2,mm W, and on EgoHumans it succeeds on only 41.1% of cases. Human-joint SE(3) is substantially stronger. Nevertheless, Shot3R has the lowest aggregate W, WA, and ATE and the highest IDF1 on both complete test sets. We interpret this as a more balanced result, not as universal dominance of every metric or viewpoint stratum.

Table 18 uses 100,000 paired bootstrap draws, resampling the 43 EgoBody recordings or 27 EgoHumans captures. Human-joint SE(3) significantly improves R0 on both datasets, while its W difference from Shot3R includes zero. This is why the main paper reports the lowest aggregate value without claiming a statistically significant universal W advantage over this strongest geometric control.

The predefined viewpoint strata are reported without result-dependent subset selection in Tables 19 and 20. Human-joint SE(3) is lower in W for EgoBody small/extreme and EgoHumans large/extreme strata; Shot3R retains lower EgoHumans ATE in every listed stratum and the best aggregate joint result. Figure 9 visualizes the complete W trend.

The fitting diagnostics in Tables 21 and 22 expose robustness that is not visible from conditional successful fits. On EgoHumans, human-joint SE(3) fails on 16/90 cases, while scene registration fails on 53/90; among four-person cases, the rates rise to 34.1% and 70.5%. The median scene-correspondence inlier ratio is 0.055 on EgoHumans, consistent with weak cross-shot overlap. The post-seal catastrophic diagnostic in Figure 10 uses a 1,m translation or 30° rotation threshold only for analysis and never selects a Test transform.

All shared rigid-transform controls leave within-camera body accuracy unchanged: local MPJPE/MPVPE remain 70.9/79.0,mm on EgoBody and 108.5/127.8,mm on EgoHumans. The fitting times in Table 21 cover only the additional geometric fit and are therefore not treated as an end-to-end runtime comparison with Shot3R.

Table 17: Complete aggregate results for the traditional geometric registration controls. W/WA are in mm and ATE in m. Fit is the success rate of the separate geometric fit; failed fits revert exactly to R0 and remain in the fixed denominator.

Dataset	Method	Access	W ↓	WA ↓	ATE ↓	IDF1 ↑	Cov. ↑	Fit ↑
EgoBody	Human3R per-shot reset	Independent shots	752.6	589.4	1.141	0.834	0.981	–
	+ pelvis translation	Boundary-time	500.2	431.8	0.068	0.896	0.981	100.0%
	+ human-joint SE(3)	Boundary-time	261.5	215.0	0.019	0.896	0.981	100.0%
	+ scene FPFH+ICP	Offline full shots	733.2	496.9	0.020	0.834	0.981	100.0%
	Shot3R	Streaming	258.8	201.7	0.016	0.985	0.981	–
EgoHumans	Human3R per-shot reset	Independent shots	1248.1	491.3	4.411	0.455	0.616	–
	+ pelvis translation	Boundary-time	1242.1	472.0	3.326	0.506	0.616	100.0%
	+ human-joint SE(3)	Boundary-time	897.0	339.9	1.731	0.510	0.616	82.2%
	+ scene FPFH+ICP	Offline full shots	1372.8	489.3	3.847	0.455	0.616	41.1%
	Shot3R	Streaming	866.8	336.9	1.423	0.571	0.616	–

EgoBody uses recording-macro aggregation over 43 recordings. EgoHumans retains the fixed case-macro headline over 90 cases; all paired intervals resample its 27 captures. Traditional controls receive annotated boundaries, while Shot3R uses its online detector.

Table 18: Paired gains with 95% clustered bootstrap intervals. Positive values versus R0 indicate lower error for the geometric control; positive values versus Shot3R indicate lower error for Shot3R.

Dataset	Control	ΔW vs. R0	ΔATE vs. R0	Resampling unit
EgoBody	Pelvis translation	252.4 [202.9, 304.3]	1.072 [0.946, 1.202]	43 recordings
EgoBody	Human-joint SE(3)	491.1 [431.9, 549.9]	1.122 [0.992, 1.254]	43 recordings
EgoBody	Scene FPFH+ICP	19.4 [−95.5, 124.8]	1.121 [0.993, 1.252]	43 recordings
EgoHumans	Pelvis translation	14.0 [−48.8, 82.2]	1.141 [0.901, 1.419]	27 captures
EgoHumans	Human-joint SE(3)	309.5 [174.1, 442.4]	2.806 [2.002, 3.696]	27 captures
EgoHumans	Scene FPFH+ICP	−107.0 [−207.0, −24.3]	0.491 [0.272, 0.734]	27 captures

Dataset	Control	ΔW vs. Shot3R	ΔATE vs. Shot3R	Resampling unit
EgoBody	Pelvis translation	241.3 [184.8, 298.0]	0.052 [0.038, 0.067]	43 recordings
EgoBody	Human-joint SE(3)	2.6 [−40.4, 53.8]	0.003 [0.000, 0.005]	43 recordings
EgoBody	Scene FPFH+ICP	474.4 [373.4, 585.7]	0.003 [0.001, 0.006]	43 recordings
EgoHumans	Pelvis translation	320.3 [191.1, 452.7]	1.989 [1.552, 2.471]	27 captures
EgoHumans	Human-joint SE(3)	24.9 [−102.0, 134.0]	0.324 [−0.197, 0.846]	27 captures
EgoHumans	Scene FPFH+ICP	441.4 [270.9, 620.5]	2.638 [1.986, 3.347]	27 captures

Table 19: EgoBody geometric controls in the three predefined viewpoint strata. W/WA are in mm and ATE in m.

Stratum	Method	W ↓	WA ↓	ATE ↓
Small ($N = 43$)	Human3R per-shot reset	546.4	423.2	0.606
	+ pelvis translation	265.3	216.3	0.047
	+ human-joint SE(3)	194.1	148.9	0.018
	+ scene FPFH+ICP	413.9	287.7	0.017
	Shot3R	201.4	160.6	0.017
Medium ($N = 43$)	Human3R per-shot reset	697.5	557.8	1.061
	+ pelvis translation	517.9	443.5	0.073
	+ human-joint SE(3)	249.1	203.6	0.020
	+ scene FPFH+ICP	650.1	451.9	0.019
	Shot3R	220.8	171.2	0.017
Extreme ($N = 43$)	Human3R per-shot reset	1013.9	787.2	1.756
	+ pelvis translation	717.3	635.5	0.086
	+ human-joint SE(3)	341.2	292.6	0.020
	+ scene FPFH+ICP	1135.7	751.1	0.023
	Shot3R	354.3	273.3	0.016

Table 20: EgoHumans geometric controls in the four predefined viewpoint strata. W/WA are in mm and ATE in m. Failure is the fitting-failure rate before the mandatory R0 fallback.

Stratum	Method	W ↓	WA ↓	ATE ↓	Failure ↓
Small ($N = 24$)	Human3R per-shot reset	1093.6	499.3	2.113	—
	+ pelvis translation	1158.9	442.2	1.355	0.0%
	+ human-joint SE(3)	872.3	351.0	0.870	16.7%
	+ scene FPFH+ICP	1136.2	497.9	1.953	75.0%
	Shot3R	850.9	357.1	0.610	—
Medium ($N = 22$)	Human3R per-shot reset	1271.4	513.1	4.559	—
	+ pelvis translation	1262.3	505.9	3.724	0.0%
	+ human-joint SE(3)	986.1	345.9	1.584	13.6%
	+ scene FPFH+ICP	1381.3	518.3	3.964	59.1%
	Shot3R	801.2	341.5	1.301	—
Large ($N = 22$)	Human3R per-shot reset	1303.0	449.5	5.031	—
	+ pelvis translation	1242.1	455.6	3.693	0.0%
	+ human-joint SE(3)	928.2	346.7	1.685	22.7%
	+ scene FPFH+ICP	1660.0	456.8	4.193	40.9%
	Shot3R	930.7	331.1	1.529	—
Extreme ($N = 22$)	Human3R per-shot reset	1338.5	502.6	6.153	—
	+ pelvis translation	1312.6	486.9	4.712	0.0%
	+ human-joint SE(3)	803.5	314.9	2.864	18.2%
	+ scene FPFH+ICP	1335.2	483.6	5.451	59.1%
	Shot3R	886.1	315.9	2.326	—

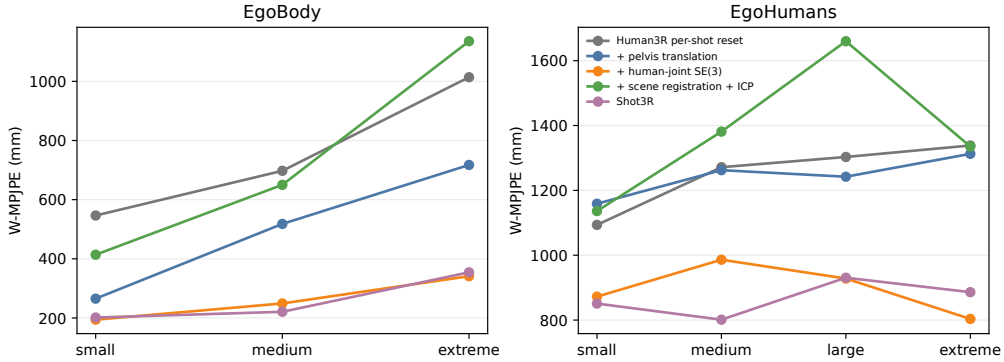


Figure 9: W-MPJPE across the predefined viewpoint strata. Every point uses the fixed denominator; failed geometric fits fall back to R0 rather than being removed. Human-joint SE(3) is a strong control and is lower in some strata, while Shot3R gives the lowest full-set aggregate W on both datasets.

Table 21: Fitting robustness and fitting-only runtime. Catastrophic denotes a post-seal diagnostic above 1,m translation or 30° rotation error; it is never used to select or reject a Test transform.

Dataset	Control	Fit failure	Catastrophic	Fit time: mean / median / p90 (s)
EgoBody	Pelvis translation	0/129 (0.0%)	112/129 (86.8%)	0.002 / 0.002 / 0.003
EgoBody	Human-joint SE(3)	0/129 (0.0%)	30/129 (23.3%)	0.078 / 0.076 / 0.087
EgoBody	Scene FPFH+ICP	0/129 (0.0%)	57/129 (44.2%)	0.524 / 0.407 / 0.955
EgoHumans	Pelvis translation	0/90 (0.0%)	90/90 (100.0%)	0.007 / 0.007 / 0.012
EgoHumans	Human-joint SE(3)	16/90 (17.8%)	74/90 (82.2%)	0.119 / 0.114 / 0.149
EgoHumans	Scene FPFH+ICP	53/90 (58.9%)	88/90 (97.8%)	39.132 / 35.422 / 76.523

Table 22: EgoHumans fitting-failure rate by number of visible people.

People	Cases	Pelvis translation	Human-joint SE(3)	Scene FPFH+ICP
2	14	0.0%	0.0%	100.0%
3	32	0.0%	3.1%	25.0%
4	44	0.0%	34.1%	70.5%

Table 23: Boundary and cross-shot diagnostics for geometric registration. Translation, post-transition root, and seam-root errors are in m; rotation error is in degrees.

Dataset	Method	Boundary trans. ↓	Boundary rot. ↓	Post root ↓	Seam root ↓
EgoBody	Human3R per-shot reset	3.374	55.7	1.364	1.332
	+ pelvis translation	2.986	55.7	0.852	0.761
	+ human-joint SE(3)	1.029	17.9	0.360	0.241
	+ scene FPFH+ICP	2.238	44.7	1.324	1.320
	Shot3R	0.679	8.0	0.364	0.309
EgoHumans	Human3R per-shot reset	8.901	104.5	3.854	2.653
	+ pelvis translation	9.905	104.5	3.777	2.312
	+ human-joint SE(3)	5.676	48.8	2.615	1.072
	+ scene FPFH+ICP	8.970	102.4	4.253	3.150
	Shot3R	4.135	23.1	2.408	1.227

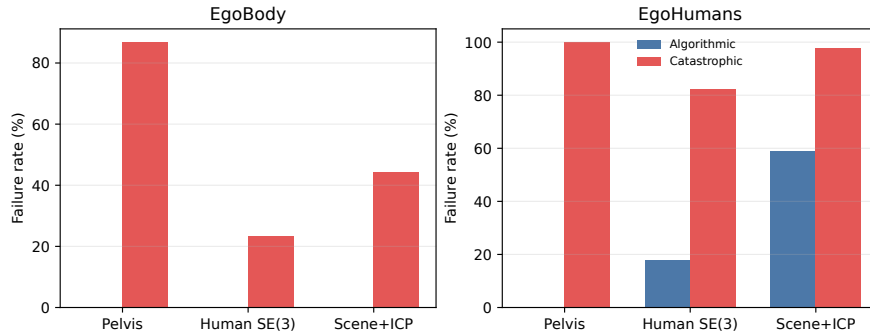


Figure 10: Algorithmic fitting failures and post-seal catastrophic geometric errors for the registration controls. Catastrophic thresholds are diagnostic only and are not used for model selection, Test-time fitting, or fallback.

D.4 EGOBODY

The main paper reports the overall EgoBody results and principal mechanism comparison. Here we provide local and boundary measures for the controlled alignment–state-propagation comparison, paired confidence intervals, cross-shot continuity, and viewpoint-stratified results.

The two aligned controls use the same checkpoint, annotated boundary, and history-conditioned alignment, disable explicit association and shared translation, and differ only in the state propagated afterward. Their equal first-post camera errors verify that the boundary alignment is held fixed. Local human accuracy is nearly unchanged; reinitialization lowers the first-post root error, whereas continuation lowers the subsequent root error.

The first detector response occurs at the true shot transition in all 129 EgoBody clips, with no early, late, or missed transitions. We report this result in text rather than repeating it in a one-row table.

Table 24: Local and boundary measures when varying the state propagated after alignment on EgoBody. Human errors are in mm, camera rotation in degrees, and the remaining camera/root errors in m.

State propagated	Local human		First post-transition			Post root
	MPJPE	PA-MPJPE	Cam. t	Cam. R	Root	
Continued	71.18	43.80	0.807	8.02	0.798	0.430
Reinitialized	70.94	43.87	0.807	8.02	0.538	0.554

Table 25: Paired Human3R–Shot3R analysis over 43 EgoBody recordings. The interval is a paired bootstrap and positive gain favours Shot3R.

Metric	Mean gain $\uparrow$	95% CI	p -value
W-MPJPE (mm)	62.3	[21.2, 111.3]	0.003708
WA-MPJPE (mm)	31.7	[15.1, 49.2]	0.000639
ATE-Sim3 (m)	0.108	[0.056, 0.173]	$< 10^{-6}$
IDF1	0.114	[0.076, 0.155]	$< 10^{-6}$

Table 26: Local reconstruction and temporal stability on EgoBody. Human errors and acceleration are in mm, ATE-Sim3 in m, and foot sliding in cm.

Method	MPJPE	PA-MPJPE	MPVPE	ATE-Sim3	Accel.	Foot sliding
Human3R	71.2	43.8	79.3	0.215	34.3	2.6
Shot3R	70.9	43.9	79.0	0.207	28.0	2.2

Table 27: Camera and human continuity at EgoBody shot transitions. Translation and root errors are in m; rotation errors are in degrees.

Method	Boundary t	Boundary R	Inter-shot t	Inter-shot R	Cross-shot root
Human3R	0.840	12.3	0.703	6.3	0.524
Shot3R	0.679	8.0	0.501	2.4	0.309

Table 28: EgoBody results by viewpoint change. W/WA are in mm and camera error is in m.

Viewpoint change	Method	W	WA	Camera	IDF1
Small	Human3R	223.4	182.1	0.048	0.965
	Shot3R	201.4	160.6	0.017	0.990
Medium	Human3R	234.9	196.2	0.072	0.888
	Shot3R	220.8	171.2	0.017	0.981
Extreme	Human3R	505.1	321.8	0.253	0.758
	Shot3R	354.3	273.3	0.016	0.983

D.5 EGOHUMANS

The viewpoint breakdown in Table 29 shows that the main gains arise at large viewpoint changes; not every metric improves in the small-view subset.

D.5.1 RECURRENT-STATE CARRY-OVER

World-frame reconstruction error alone cannot distinguish losing a fixed inter-shot coordinate relation from changing the trajectory within the new shot. We isolate the latter on a fixed 90-case EgoHumans protocol using the same original Human3R checkpoint. The continuous setting processes both shots in order and carries state across the transition; the reinitialized setting processes the identical 50 frames after the transition from the initial state. Both use the same preprocessing, and no Shot3R module is enabled.

Each EgoHumans shot is recorded by a static camera, so the ground-truth relative camera motion within the second shot is the identity. We anchor the first prediction after the transition and measure

$$\begin{aligned}\Delta\hat{C}_k &= \hat{C}_{B0}^{-1}\hat{C}_{Bk}, \\ d_T(k) &= \left\| \text{trans}(\Delta\hat{C}_k) \right\|_2, \\ d_R(k) &= \cos^{-1} \left(\frac{\text{tr}(\text{Rot}(\Delta\hat{C}_k)) - 1}{2} \right).\end{aligned}\tag{11}$$

Any fixed world-frame transform between shots cancels in this relative pose, so the measure does not penalize reinitialization for lacking the inter-shot coordinate relation. We average offsets $k = 1, \dots, 16$ within each case, then average cases within a capture and give all 27 captures equal weight.

As Table 31 shows, continued state produces 1.124 m/14.58° mean translation/rotation drift, compared with 0.026 m/0.20° after reinitialization. Reinitialization is lower on all 27 captures for both measures. The capture-paired mean reductions and 95% bootstrap intervals are 1.098 [0.832, 1.372] m and 14.38° [8.92°, 20.49°]; both one-sided paired Wilcoxon tests give $p = 7.45 \times 10^{-9}$.

The conclusion is stable to the measurement window. For windows of 10/32/49 frames, continued versus reinitialized translation drift is 1.045/0.019, 1.256/0.039, and 1.387/0.047 m; rotation drift is 14.34°/0.16°, 15.12°/0.27°, and 15.65°/0.34°. The ordering is also unchanged in all four fixed viewpoint strata. This control does not claim that reinitialization alone solves multi-shot reconstruction—it loses the inter-shot coordinate relation. Instead, it shows that the preceding-shot state causes time-varying error within the subsequent shot, supporting separate treatment of alignment and recurrent-state propagation.

The ablation in Table 32 confirms the same pattern as EgoBody. Reinitialization alone breaks world-frame continuity; camera alignment restores the inter-shot camera relation, while the full method also uses person association and the shared transform to improve human trajectories and identity continuity.

Table 29: EgoHumans results by viewpoint change. W/WA are in mm.

Method	Viewpoint stratum	N	W-MPJPE $\downarrow$	WA-MPJPE $\downarrow$	IDF1 $\uparrow$	Coverage $\uparrow$
Human3R	small	24	767.7	331.5	0.559	0.658
Shot3R	small	24	850.9	357.1	0.593	0.657
Human3R	medium	22	824.7	345.9	0.414	0.637
Shot3R	medium	22	801.2	341.5	0.572	0.605
Human3R	large	22	1221.1	388.6	0.501	0.647
Shot3R	large	22	930.7	331.1	0.558	0.622
Human3R	$\geq 150^\circ$	22	1228.4	400.0	0.404	0.611
Shot3R	$\geq 150^\circ$	22	886.1	315.9	0.559	0.576

The four fixed strata contain 24, 22, 22, and 22 clips. W/WA are available for all clips in this internal comparison; IDF1 and coverage retain the stratum denominator.

Table 30: Paired Human3R–Shot3R analysis on 90 EgoHumans clips from 27 captures. Intervals resample captures and then clips within each capture.

Metric	Mean gain $\uparrow$	95% CI	p -value
W-MPJPE (mm)	138.3	[0.5, 322.9]	0.035713
WA-MPJPE (mm)	28.9	[-6.8, 68.7]	0.057267
ATE-SE3 (m)	0.515	[0.258, 0.925]	$< 10^{-6}$
IDF1	0.100	[0.046, 0.152]	0.000799
Coverage	-0.023	[-0.058, 0.008]	0.122279

Table 31: Isolating recurrent-state carry-over on the fixed 90-case EgoHumans protocol. The same original Human3R checkpoint processes identical frames after the transition with the preceding-shot state or a reinitialized state. We report mean within-shot camera drift over the next 16 frames after anchoring the first pose, with equal weight for each of 27 captures.

State at the shot transition	Mean trans. drift (m) $\downarrow$	Mean rot. drift ($^\circ$) $\downarrow$
Continued	1.1237	14.578
Reinitialized	0.0258	0.200

Table 32: Effects of alignment, state reinitialization, and association on EgoHumans. W/WA/MPJPE are in mm and cross-shot root error is in m.

Method	W	WA	MPJPE	Cross-shot root	IDF1	Coverage
Human3R	1005.1	365.7	107.6	1.723	0.471	0.639
State reinitialization	1248.1	491.3	108.5	2.653	0.455	0.616
Camera alignment	1072.4	370.1	108.5	2.113	0.455	0.616
Camera alignment + association	1058.5	372.6	109.9	2.113	0.556	0.600
Shot3R	866.8	336.9	108.5	1.227	0.571	0.616

D.6 HARMONY4D

Harmony4D uses the same model and inference settings as the other datasets. The complete method comparison appears in Table 12; Table 33 adds only local body errors and translation error at the shot transition.

D.6.1 SHARED-TRANSLATION WEIGHT

We study the shared-translation weight on 12 Harmony4D training sequences reserved for development, nine of which provide all required metrics. Local MPJPE, MPVPE, IDF1, and Coverage do not change with this output-space weight and are omitted rather than repeated. We use $\lambda = 0.5$ across all datasets. On this development split, it achieves the lowest mean W and WA while retaining competitive camera ATE and cross-shot root continuity, providing a balanced combination of camera- and person-derived alignment.

D.6.2 CROSS-SHOT PERSON ASSOCIATION

Among 88 clips, 84 contain at least one valid person match. Of 147 valid matches, 142 are correct, giving 96.60% pair accuracy and 80.68% identity continuation. The ground-truth-identity row in Table 35 changes only the identity labels and is included as an upper bound.

We further vary only the geometric cues in the matching cost, while retaining the same frozen predictions, per-case median normalization, and Hungarian assignment. Each correspondence is fixed before ground-truth identities are opened and used only for scoring.

Pelvis position already gives high pair accuracy. Adding torso orientation and root-centered joint structure further increases the number of correct pairs, identity continuation, and IDF1. This supports the complementarity of the three predicted geometric cues, while association remains a lightweight supporting mechanism after inter-shot alignment rather than the central representation-learning contribution.

D.6.3 THREE-SHOT CONTROL

Four additional Harmony4D sequences contain three shots and two transitions. Shot3R improves W, WA, IDF1, and cross-shot human continuity, while ATE-Sim3 is slightly higher than Human3R. OnlineHMR completes inference on all four streams, but its 0.121 Coverage leaves only three with evaluable W/WA. Its lower W/WA must therefore be read together with this count; across all four streams, Shot3R has lower camera error and higher IDF1 and Coverage. These four sequences provide a limited repeated-application check rather than a benchmark-scale evaluation.

Table 33: Additional diagnostics on 88 Harmony4D clips. Local body errors are in mm and transition-translation error is in m.

Method	MPJPE ↓	MPVPE ↓	Boundary trans. ↓
Human3R	98.3	116.7	0.864
Shot3R (ours)	97.9	116.1	1.042

Table 34: Sensitivity to the shared-translation weight on the Harmony4D development split. W/WA are in mm; ATE and cross-shot root error are in m.

λ	W mean ↓	WA mean ↓	ATE ↓	Cross-shot root ↓
0.00	630.4	322.7	0.020	1.175
0.25	536.2	299.1	0.019	0.857
0.50 (used)	499.4	281.5	0.017	0.585
0.75	517.9	288.2	0.016	0.507
1.00	606.5	321.5	0.015	0.560

Table 35: Cross-shot person association on Harmony4D. Pair accuracy uses 147 evaluator-valid endpoint pairs, whereas identity continuation uses all 176 ground-truth continuations. The ground-truth-identity row changes only identity labels and serves as an evaluation upper bound.

Measure	Result
Clips with at least one valid match	84 / 88
Pair accuracy	142 / 147 (96.60%)
Clip-level accuracy (95% CI)	94.05% [88.10, 98.81]
Identity continuation	142 / 176 (80.68%)
Framewise IDF1 (95% CI)	63.61% [59.33, 67.99]
Ground-truth identity upper bound (95% CI)	69.41% [65.76, 73.01]

Table 36: Association-cue ablation on 88 Harmony4D cases. Pair accuracy uses evaluator-valid endpoint matches for each configuration, identity continuation uses all 176 cross-shot ground-truth continuations, and IDF1 is averaged over the 88 cases.

Association cues	Correct / eval.	Pair acc.	ID continuation	IDF1
Pelvis	138 / 144	95.83%	78.41%	0.620
Pelvis + torso	139 / 144	96.53%	78.98%	0.630
Pelvis + torso + root-centered joints	142 / 147	96.60%	80.68%	0.635

Table 37: Harmony4D three-shot results. W/WA are in mm; camera and root errors are in m. Parentheses give the number of evaluable sequences; Coverage is computed over all four streams.

Method	W	WA	ATE-Sim3	IDF1	Coverage	Cross-shot root
Human3R	2091.9	719.4	1.618	0.377	0.936	4.539
Shot3R	1304.8	651.2	1.700	0.434	0.936	1.725
OnlineHMR	1065.7 (3)	304.3 (3)	3.393	0.055	0.121	–

D.7 FULL QUALITATIVE COMPARISON

The main paper enlarges the four comparisons most directly related to our claim. Figure 11 retains all seven evaluated methods on the same three transitions. Each column shows the method’s available native output; unavailable camera or scene geometry is not imputed.

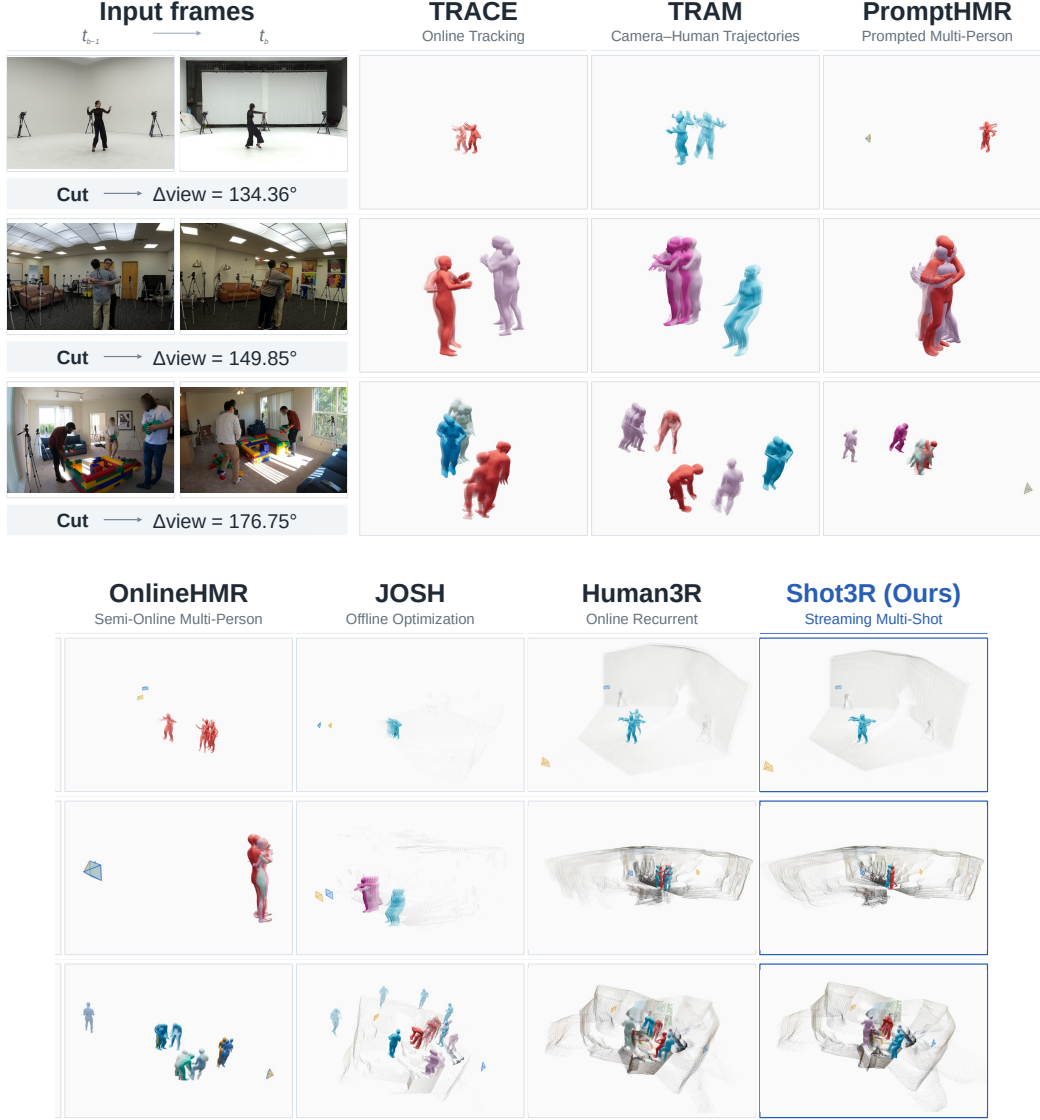


Figure 11: **Complete qualitative comparison.** TRACE, TRAM, PromptHMR, OnlineHMR, JOSH, Human3R, and Shot3R are shown on the same inputs. Viewpoint changes are 134.36° , 149.85° , and 176.75° from top to bottom; blue borders mark Shot3R.

Part III: Scope and Limitations

E LIMITATIONS

Our evaluation focuses on shot transitions caused by viewpoint changes within the same physical scene. Maintaining spatial and identity consistency under complex editing, repeated entrances and exits, rapid visual changes, and very long videos remains challenging.

Streaming and offline methods have different temporal access, which is stated throughout the comparisons. Human errors computed only on valid matches should also be read together with Coverage. Improved camera or identity consistency does not imply improvement in every body metric; the complete Harmony4D results retain these metric-level differences.